Supplementary material

Separating decision-rule failures from representation failures in external validation of diabetic retinopathy grading models

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Table S1: Analysis plans and the runs they cover. Commit identifiers and times (UTC) are from the public repository history; launch times are from the Kaggle kernel listing archived in docs/generated/paper/kaggle_runs.csv. Every run listed was launched after the commit of its plan (for the post-hoc checks, of its code) and pinned the code commit it ran. The two outcome plans are the analysis-plan files in docs/ of the repository; the second-stage plan was written after an internal pre-submission review of an earlier draft.

Plan	Commit	Committed (UTC)	First run launched (UTC)
Repository history begins (evaluation protocol present)	`5846b1b`	2026-08-31 17:51:25	–
Part A: derivation test of the macular-oedema grade	`f79abc4`	2026-09-07 17:37:44	–
CORAL, CORN and ConvNeXt interventions (hypotheses in EXPERIMENTS.md)	`b723945`	2026-09-05 17:10:36	2026-09-05 17:10:42
First-stage outcomes O1–O9 (first-stage plan)	`46f1bdd`	2026-10-16 11:03:04	2026-10-16 11:03:32
Second-stage outcomes A1–E2 (second-stage plan)	`1456815`	2026-10-20 11:21:07	2026-10-20 11:21:57
Post hoc: acquisition statistics (code commit)	`d3c42f7`	2026-10-21 07:10:13	2026-10-21 07:10:59
Post hoc: overlap audit of DeepDRiD and SUSTech-SYSU (code commit)	`fe82a52`	2026-10-21 07:16:26	2026-10-21 07:16:41

Table S2: Every pre-specified outcome. Prediction and falsifying result as committed in the plan (first stage: commit `46f1bdd`, 2026-10-16 11:03:04 UTC; second stage: commit `1456815`, 2026-10-20 11:21:07 UTC; Table S1), the result, and the verdict. Descriptive outcomes carried no prediction. The last block lists the analyses added after an internal editorial assessment of the manuscript; they were not pre-specified.

Outcome	Prediction (falsified if)	Result	Verdict
First stage (seeds, reverse direction, DDR, five-fold RETFound)			
O1 noise floor	Descriptive: s.d. over five seeds of the reference model	development 0.0043, APTOS 0.0026, DDR 0.0116 (QWK)	measured
O2 spurious per-grade effects	More than 5 of 50 seed-pair per-grade recall comparisons on APTOS have an interval excluding zero at R1 ($\leq$5)	15 of 50	held
O3 mild collapse across seeds	Every seed: APTOS mild recall below 15% at R1 and above 50% at R4m (any seed outside)	R1 5.4–6.2%, R4m 63.8–72.7%	held
O4 reverse collapse	Messidor-2 mild recall below 20% at R1 ($\geq$20%)	34.4%	falsified
O5 reverse recovery	R4m mild recall at least three times R1 and at least 40% (either fails)	44.4% against 34.4%	falsified
O6 reverse under-grading	Mean predicted-minus-true grade below zero on Messidor-2 at R1, mild errors mostly to no DR (over-grades)	−0.26; mild to no DR 175, to moderate 2	held
O7 rule versus model	Median within-model range over R1–R3 exceeds the between-model range at R1, for mild recall and referable sensitivity (between-model range larger)	APTOS mild 5.7 vs 12.7, sensitivity 0.9 vs 1.3; DDR mild 21.0 vs 24.8, sensitivity 7.1 vs 11.4 points	falsified
O8 threshold transfer to DDR	The development threshold for 85% sensitivity lands more than 5 points from target on DDR (within ±5)	63.38%	held
O9 RETFound at five folds	Matched development QWK of RETFound minus DenseNet121-224 has an interval below zero (contains zero or positive)	−0.0414 [−0.0583, −0.0238]	held
Second stage (mechanism, label-free rules, sample size, new data)			
A1 conditional shift	E[$s \mid y$] above development for at least 3 of 5 grades in at least 8 of 10 models on APTOS, below on DDR (either fails). The overall direction on APTOS and DDR was known; DeepDRiD and SUSTech-SYSU were added without a predicted direction	APTOS 10, DDR 10 of 10 models; DeepDRiD below in 10, SUSTech-SYSU above in 10	held

Outcome	Prediction (falsified if)	Result	Verdict
A2 degradation	At least one degradation lowers the mean predicted-minus-true grade by $\geq$0.10 and referable sensitivity by $\geq$10 points (none does both)	JPEG quality 30: −0.45 grade and −36.8 points (reference model), −0.53 and −45.6 (selected)	held
A3, A4 image statistics, grade mix	Descriptive	Table S8, Table 1	–
B label-free rule	R5 closes at least 25% of the DDR referral-sensitivity gap between R1 and R4 in at least 6 of 10 models (fewer)	2 of 10	falsified
C sample size	With 200 labelled images the probability of a held-out sensitivity below 80% is under 20% on every external set (any set at or above)	9.4–15.8% on sets with 40–45% referable images	held
D descriptive additions	Descriptive	Results; Table S9; rule matrices (Tables S16–S19)	–
E1 two further external sets	Inference only, no prediction	DeepDRiD and SUSTech-SYSU scored by every development model	done
E2 large training set	On at least 2 of 6 external sets the EyePACS-fitted threshold for 85% misses by more than 5 points while referral AUC stays $\geq$0.90 (no set misses)	6 of 6 sets for both models (90.8–100.0%); AUC 0.938–0.999	held
Added after an internal editorial assessment (post hoc)			
Sample size in referable cases	–	Table S10, Fig. 6	post hoc
Label-free rules against argmax	–	Table S11	post hoc
Case mix and acquisition	–	Table S12	post hoc
Kendall τ intervals; like-for-like mild F1	–	Results	post hoc

Table S3: Reporting checklist (TRIPOD+AI and CLAIM). Each reporting domain mapped to the place in the article where it is addressed; the journal's checklist forms are completed from this mapping at submission.

Domain	Where and how it is reported
Title and abstract	Title names the task and the finding; the abstract states data, models, external sets, decision rules and main results.
Background and objectives	Introduction, including what this study adds to ORDER-DR and to recent calibration studies.
Data sources	Section 2.2 and Table 1: IDRiD, Messidor-2 (adjudicated grades), EyePACS, APTOS 2019, DDR, DeepDRiD, SUSTech-SYSU.
Participants and eligibility	Every gradable image with a DR grade 0–4; DDR grade 5 (ungradable) excluded. No patient-level demographics are released by the corpora (limitation).
Data preparation	Section 2.3 and Table S4: retinal crop, cache at 560 pixels, resizing, augmentation.
Outcome (reference standard)	International 5-grade DR scale; referable DR is grade $\geq$2, defined by DR alone because macular oedema is not graded externally.
Sample size	Fixed by the public corpora; bootstrap intervals throughout; the local recalibration sample size is an explicit result (Results; Fig. 6 and Table S10).
Missing data	Messidor-2 has no three-grade DME label; the DME loss is masked for those images.
Data partitions	Frozen 5-fold split of the development pool; external sets never used for training, selection or threshold fitting except in the diagnostic R4/R4m; content audit (Note 7).
Model specification	Section 2.3, Table 2 and Table S4.
Model output and decision rule	Section 2.4 and Note 1 (R1–R5b).

Domain	Where and how it is reported
Performance measures	QWK, Cohen's κ, accuracy, per-grade recall, precision and F1, referable sensitivity and specificity, per-threshold AUC, Harrell's C, Brier score, ECE, PPV and NPV.
Uncertainty	Paired bootstrap (2 000 resamples) over patients where identified, otherwise images.
Class imbalance	Grade-balanced sampler; per-grade results for every grade.
Fairness	No subgroup attributes are released; between-population behaviour is studied across four external populations and two directions.
Analysis plans	Committed to version control before each analysis (Table S1); every outcome with its prediction, falsifying result and verdict in Table S2; post-hoc analyses labelled.
Code and data availability	Data and code availability statement.
Ethics	Ethics statement (public, de-identified data).
Use of AI tools	Section 2.8 (AI-assisted coding) and the generative-AI declaration (editing of the text).
Limitations	Discussion.

Table S4: Configurations. Each configuration changes one factor relative to its control. Dev QWK: development out-of-fold DR QWK under the matched decision rule (fold-wise cross-fitted cut-points); folds: development folds trained.

Model	Control	Backbone	DR head	Pixels	Pretraining	Epochs	Factor	Dev QWK
E08	–	DenseNet121	cumulative	448	ImageNet + EyePACS (5 ep)	40	reference	0.8646
E09	E08	DenseNet121	cumulative	224	ImageNet + EyePACS	40	resolution	0.8389
E10	E08	DenseNet121	cumulative	640 (560 effective)	ImageNet + EyePACS	40	resolution	0.8749
E17NAT	E10	DenseNet121	cumulative	640	ImageNet + EyePACS	40	native-resolution source files	0.8662
E11FULL	E10	EfficientNet-B3	cumulative	448	ImageNet + EyePACS	30	backbone (selected model)	0.8954
E14MAC	E08	DenseNet121	cumulative	448 + macula crop	ImageNet + EyePACS	40	fovea-centred crop for DME head	0.8595
E15LPFT	E08	DenseNet121	cumulative	448	ImageNet + EyePACS	40 (LP 5)	linear probe then fine-tune	0.8655
E20CORAL	E08	DenseNet121	CORAL	448	ImageNet + EyePACS	40	ordinal head	0.8569
E22CORN	E08	DenseNet121	CORN	448	ImageNet + EyePACS	40	ordinal head	0.8564
E21CNXA	E08	ConvNeXt-tiny	cumulative	448	ImageNet + EyePACS	40	backbone (2 folds)	0.8267 (2 f)
I24FULL	E09	RETFound ViT-L/16	cumulative	224	RETFound (colour fundus)	30 (LP 5)	foundation encoder	0.7976
S1E08–S4E08	E08	DenseNet121	cumulative	448	ImageNet + EyePACS	40	seed only	–
R1APTOS	E08	DenseNet121	cumulative	448	ImageNet + EyePACS	40	trained on APTOS (reverse)	–

Table S5: Pre-specified interventions, shipped and matched decision rule. Difference (intervention minus control) in development DR QWK with 95% paired-bootstrap intervals (DME QWK for the macula crop; APTOS DR QWK for the ensemble).

Intervention	Control	Folds	Matched difference [95% CI]	Verdict
EfficientNet-B3 backbone	E10	5	+0.0207 [+0.0105, +0.0320]	confirmed
Macula-centred crop (DME QWK)	E08	5	−0.0237 [−0.0566, +0.0094]	falsified
Linear probe then fine-tune	E08	5	+0.0008 [−0.0105, +0.0126]	falsified
Native 640-px resolution	E10	5	−0.0086 [−0.0199, +0.0025]	falsified
CORAL ordinal head	E08	5	−0.0076 [−0.0184, +0.0032]	falsified
CORN ordinal head	E08	5	−0.0082 [−0.0207, +0.0036]	falsified
ConvNeXt-tiny backbone	E08	2	−0.0362 [−0.0593, −0.0153]	falsified, significantly worse
RETFound ViT-L/16 (fine-tuned)	E09	5	−0.0414 [−0.0583, −0.0238]	falsified, significantly worse
Six-model ensemble (APTOS, external)	E10	5	+0.0025 [−0.0016, +0.0066]	falsified

Table S6: Decision-rule reanalysis of a published model family (ORDER-DR, APTOS→Messidor-2). Mean ± s.d. over the authors' three seeds, computed from their released image-level probabilities on 1,744 Messidor-2 images (270 mild). For ORDER-DR the R2 row uses the authors' own fusion and cut-points and reproduces their reported result. R4/R4m use external labels (two-fold cross-fitting) and are diagnostic bounds, not performance claims.

Model	Rule	QWK	Mild rec. (%)	Mild F1	Sens. (%)	Spec. (%)
CB-384	R1 argmax	0.500 ± 0.018	2.0	0.027	32.8	99.5
	R2 val.-fitted	0.544 ± 0.029	2.5	0.035	38.2	99.0
	R4 ext. QWK	0.672 ± 0.030	27.8	0.231	55.1	95.4
	R4m ext. recall	0.614 ± 0.060	38.4	0.255	72.6	85.2
LORS-384	R1 argmax	0.531 ± 0.014	3.5	0.049	36.3	98.9
	R2 val.-fitted	0.570 ± 0.023	2.7	0.041	42.8	98.7
	R4 ext. QWK	0.683 ± 0.045	30.5	0.245	55.9	96.1
	R4m ext. recall	0.677 ± 0.046	36.5	0.285	67.6	91.3
Swin-T	R1 argmax	0.552 ± 0.045	5.2	0.062	35.4	96.9
	R2 val.-fitted	0.564 ± 0.026	3.3	0.045	43.2	96.2
	R4 ext. QWK	0.674 ± 0.022	13.0	0.147	64.8	93.8
	R4m ext. recall	0.629 ± 0.041	30.2	0.233	75.0	88.9
ORDER-DR	R1 argmax	0.509 ± 0.017	2.7	0.040	34.1	99.2
	R2 val.-fitted	0.642 ± 0.036	50.5	0.277	39.8	99.1
	R4 ext. QWK	0.676 ± 0.025	24.2	0.212	56.7	95.4
	R4m ext. recall	0.638 ± 0.049	42.0	0.284	70.0	88.8

Table S7: Models fine-tuned on EyePACS (outcome E2). Trained on four folds of a patient-grouped EyePACS split (28,096 images); the fifth fold (7,030 single-graded images) selected the checkpoint and is the development set. Referral area under the curve, mean predicted-minus-true grade and referable sensitivity and specificity at the shipped rule (R1), sensitivity with cut-points refitted on external labels (R4), and with a threshold fitted on the held-out EyePACS fold for 85% sensitivity. The development row shows the referral area under the curve on the held-out fold itself.

Model (EyePACS-trained)	Corpus	AUC	Pred.−true	Sens. R1	Spec. R1	Sens. R4	Sens. thr. 85
DenseNet121	EyePACS fold 0 (dev.)	0.917	–	–	–	–	–
	APTOS	0.941	+0.20	99.7	83.0	96.6	100.0
	DDR	0.957	−0.18	77.8	96.8	83.9	90.9
	Messidor-2	0.940	+0.09	86.9	86.5	69.6	95.2
	IDRiD	0.959	+0.15	97.5	71.5	90.1	98.8
	DeepDRiD	0.965	+0.02	94.9	87.9	88.5	98.7
	SUSTech	0.999	+0.13	99.2	98.5	98.4	100.0
EfficientNet-B3	EyePACS fold 0 (dev.)	0.917	–	–	–	–	–
	APTOS	0.938	+0.14	98.9	83.7	96.4	99.9
	DDR	0.952	−0.23	73.8	97.7	80.1	90.8
	Messidor-2	0.956	−0.08	85.3	94.3	77.2	95.8
	IDRiD	0.984	+0.06	96.6	87.0	94.4	99.4
	DeepDRiD	0.969	−0.01	92.9	89.7	91.2	98.7
	SUSTech	0.999	+0.05	98.8	98.5	99.0	100.0

Table S8: Image degradation and image statistics (outcomes A2, A3). Top: change relative to the undegraded re-scoring of the development pool (out of fold) in mean predicted-minus-true grade and in referable sensitivity at the shipped rule. Bottom: median image statistics per corpus (interquartile range in the repository).

Degradation	Reference model		Selected model	
	Δ pred.−true	Δ sens. (pts)	Δ pred.−true	Δ sens. (pts)
JPEG quality 30 at 512 px	−0.45	−36.8	−0.53	−45.6
Downsampled to 256 px	−0.15	−6.2	−0.04	−1.3
Gaussian blur, sigma 2 px	−0.30	−15.4	−0.24	−15.9
Gamma 1.8 (darker)	−0.16	−8.7	−0.09	−6.7
Contrast reduced to 60%	−0.13	−5.1	−0.06	−2.3
Corpus	**Long side (px)**	**JPEG quality**	**Brightness**	**Laplacian var.**
IDRiD	4,288	50	97	52
Messidor-2	512	95	87	39
APTOS	2,048	–	91	52
DDR	512	95	93	70
DeepDRiD	1,824	95	93	47
SUSTech	2,880	75	120	51

Table S9: What the operating point means for a screening programme. Referable patients missed and false referrals per 10 000 people screened, and predictive values, for the selected model at its shipped rule (R1) and with cut-points refitted on local labels (R4), at three referable prevalences. Sensitivity and specificity are those measured on each external set.

Corpus	Rule	Sens. (%)	Spec. (%)	Prevalence (%)	Missed /10 000	False referrals /10 000	PPV (%)	NPV (%)
APTOS	R1	99.3	84.3	5	4	1,489	25.0	100.0
		99.3	84.3	10	7	1,411	41.3	99.9
		99.3	84.3	20	15	1,254	61.3	99.8
	R4	96.5	86.6	5	17	1,275	27.4	99.8
		96.5	86.6	10	35	1,208	44.4	99.6
		96.5	86.6	20	70	1,074	64.2	99.0
DDR	R1	70.3	98.7	5	149	124	73.9	98.4
		70.3	98.7	10	297	117	85.7	96.8
		70.3	98.7	20	594	104	93.1	93.0
	R4	81.2	94.2	5	94	550	42.5	99.0
		81.2	94.2	10	188	521	60.9	97.8
		81.2	94.2	20	375	463	77.8	95.3
DeepDRiD	R1	90.3	92.3	5	49	728	38.3	99.4
		90.3	92.3	10	97	689	56.7	98.8
		90.3	92.3	20	194	613	74.7	97.4
	R4	91.4	91.1	5	43	845	35.1	99.5
		91.4	91.1	10	86	800	53.3	99.0
		91.4	91.1	20	171	711	72.0	97.7
SUSTech	R1	98.8	98.5	5	6	145	77.3	99.9
		98.8	98.5	10	12	137	87.8	99.9
		98.8	98.5	20	24	122	94.2	99.7
	R4	98.4	98.6	5	8	131	79.0	99.9
		98.4	98.6	10	16	124	88.8	99.8
		98.4	98.6	20	32	110	94.7	99.6

Table S10: How many locally labelled cases a referral threshold needs (post hoc). The threshold is set to the largest value giving at least 85% sensitivity on the referable cases of a labelled sample; harm is a population sensitivity below 80%. Top: by the number k of referable cases in the sample. The closed form is distribution-free (with continuous scores the population sensitivity of the cut at the j-th largest of k referable scores, $j = \lceil 0.85k \rceil$, follows a Beta$(j, k - j + 1)$ distribution); the simulated range covers the selected and reference models on the four external sets (2 000 draws each, sensitivity measured on the corpus's referable images not drawn; k at most a quarter of them). The last two columns give the sample sensitivity at which the cut must be set for the risk to stay at or below 5% or 10%. Bottom: by the number of consecutive images labelled at a given referable prevalence (closed form, binomial number of referable cases; no referable case counts as harm).

Referable cases k	Harm, closed form (%)	Harm, simulated (%)	Sample sens. for ≤5%	for ≤10%
20	41.1	39.9–42.2	100.0	95.0
40	28.6	26.7–30.0	92.5	90.0
60	21.3	21.1–23.8	90.0	88.3
80	16.4	15.3–19.2	88.8	87.5
100	12.9	11.3–16.4	87.0	86.0
120	10.2	9.4–13.2	86.7	85.8
160	6.6	6.2–9.0	85.6	84.4
200	4.3	4.2–7.1	85.0	84.0
300	1.6	1.4–2.4	84.0	83.3

Images labelled	Harm (%) at a referable prevalence of 5% / 10% / 20%
100	44.4 / 38.5 / 31.7
200	38.5 / 31.6 / 22.6
500	28.8 / 19.5 / 10.5
1,000	19.5 / 10.5 / 3.6
2,000	10.5 / 3.6 / 0.5

Table S11: Label-free prior adaptation against the decoding it uses (post hoc). R5 (EM prior adaptation) and R5b (black-box shift estimation) decode the adapted class posteriors by argmax, so the like-for-like reference is R0a, the argmax of the unadapted posteriors, not the 0.5 rule R1. Median (range) over the ten development models. Mild recall: median over models.

Corpus	Quantity	R1	R0a	R5	R5b	R4 (diagnostic)
APTOS	Referable sensitivity (%)	99.2 (98.2–99.5)	99.1 (98.3–99.5)	99.3 (98.9–99.7)	99.3 (99.1–99.7)	97.0 (96.3–98.6)
	QWK	0.884	0.882	0.883	0.887	0.897
	Mild recall (%)	5.4	3.0	0.0	0.0	15.1
DDR	Referable sensitivity (%)	63.9 (58.9–70.3)	64.2 (59.3–70.9)	66.4 (62.4–71.6)	67.3 (63.3–71.2)	79.4 (70.0–81.2)
	QWK	0.736	0.721	0.704	0.697	0.793
	Mild recall (%)	36.3	29.6	0.0	0.0	28.3
DeepDRiD	Referable sensitivity (%)	84.3 (77.8–90.3)	84.5 (77.0–90.9)	87.2 (78.0–92.9)	87.6 (77.7–92.8)	85.1 (76.5–91.4)
	QWK	0.793	0.782	0.775	0.767	0.821
	Mild recall (%)	24.8	20.0	0.0	0.0	32.7
SUSTech-SYSU	Referable sensitivity (%)	98.8 (97.6–99.2)	99.0 (98.0–99.4)	98.9 (98.6–99.8)	99.3 (98.8–99.8)	98.2 (97.4–99.8)
	QWK	0.920	0.916	0.912	0.918	0.933
	Mild recall (%)	39.6	33.3	0.0	0.0	20.8

Table S12: Case mix and discrimination of each population (post hoc). Areas under the ROC curve, range over the ten development models. Referral AUC standardised to the development grade mix weights each image by the development share of its grade over the corpus share, within referable and within non-referable images. Adjacent-grade AUCs use $P(y > 0)$ for no DR against mild and $P(y > 1)$ for the other pairs. Acquisition (post hoc, run X6ACQ): the largest area under the curve, taken in whichever direction separates better, with which any single acquisition statistic of the image (resolution, file size, estimated JPEG quality, share of the frame inside the retinal mask, brightness, contrast, sharpness) separates referable from non-referable images, and any DR from no DR, without a model; 0.5 is no separation. Contrast and sharpness also rise with lesions.

Population	Mild (%)	Referral AUC	Standardised	No DR vs mild	Mild vs moderate	No DR vs moderate	Acq. referable	Acq. any DR
Development	13.1	0.952–0.977	–	0.664–0.815	0.878–0.927	0.939–0.979	0.79	0.74
APTOS	10.1	0.914–0.973	0.900–0.969	0.973–0.996	0.534–0.800	0.999–1.000	0.82	0.86
DDR	5.0	0.921–0.955	0.923–0.952	0.748–0.860	0.769–0.832	0.916–0.958	0.60	0.61
DeepDRiD	11.1	0.943–0.969	0.929–0.964	0.686–0.766	0.853–0.901	0.917–0.969	0.74	0.77
SUSTech-SYSU	2.1	0.996–0.999	0.984–0.997	0.893–0.958	0.922–0.981	0.998–1.000	0.79	0.78

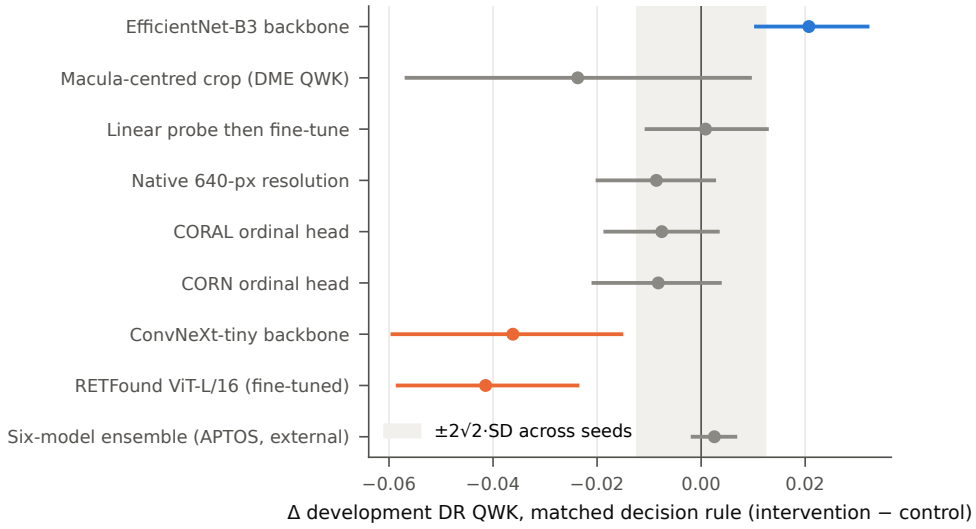


Figure S1: Pre-specified interventions on the model. Difference in development DR QWK (intervention minus control) under the matched decision rule with 95% paired-bootstrap intervals; the macula-centred crop was specified on the DME head and the ensemble on the external set. The grey band is $\pm 2\sqrt{2}$ standard deviations of matched development QWK across five seeds of the reference configuration (0.0123). Blue, improvement confirmed; orange, interval excludes zero in the unfavourable direction; grey, indistinguishable.

Supplementary Note 1. Decision rules

Every model emits, for DR, four logits whose sigmoids estimate $P(y > k)$, $k = 0, \ldots, 3$ (cumulative head and CORAL), or conditional probabilities $P(y > k \mid y > k-1)$ that are chained by cumulative product (CORN). From these the expected grade $s = \sum_k P(y > k)$ is computed.

R1 (shipped). Grade $= \#\{k : P(y > k) > 0.5\}$ after enforcing monotonicity by a running minimum.

R2 (development QWK cut-points). Four ordered cut-points on s maximising QWK on the development out-of-fold predictions, by coordinate ascent over 60 candidate values per cut-point (8 rounds), applied unchanged externally.

R3 (R2 with a referral constraint). R2 with the grade-1/2 cut-point replaced by the largest value that keeps development referable sensitivity at or above 0.90.

R4 and R4m (external, diagnostic). The same search on the external labels (80 candidates), with the external set split at random into two halves (seed 0); cut-points fitted on one half are applied to the other. R4 maximises QWK, R4m macro-averaged recall.

R5 (EM prior adaptation). Class probabilities $\pi_k(x)$ are derived from the monotone cumulative outputs ($\pi_0 = 1 - p_0$, $\pi_k = p_{k-1} - p_k$, $\pi_4 = p_3$), clipped and renormalised. The source prior is the mean development out-of-fold class probability; the target prior is estimated on the unlabelled external images by the EM procedure of Saerens *et al.*; posteriors are reweighted by the ratio of the priors and decoded by argmax.

R5b (black-box shift estimation). Importance weights $w = C^{-1}\mu$, with C the development confusion matrix of R1 and μ the external distribution of R1 predictions (negative weights set to zero); posteriors reweighted and decoded as for R5.

R4/R4m use external labels and are never reported as performance. Changing the development grid from 60 to 80 values changes the reference model's matched development QWK by 0.0033; 60 is used for all development quantities because it reproduces the archived matched comparisons exactly. Argmax decoding of the class probabilities derived from cumulative outputs rarely produces a mild prediction: R0a predicted fewer mild images than R1 on every corpus, and R5 predicted any at all in only 7 of 40 model–corpus pairs, which limits what prior adaptation can do for the mild grade whatever the prior estimate. R0a, not R1, is therefore the like-for-like reference for R5 and R5b (Table S11).

Supplementary Note 2. Macular oedema: a supervision ceiling and an underivable grade

No intervention moved the three-grade DME head. A crop centred on the fovea by a localiser validated before use (median error 0.196 disc diameters), which encodes the clinical definition of the grade, changed DME QWK by −0.0237 [−0.0566, +0.0094]. Only 51 of the 516 three-grade images (9.9%) carry the middle grade, and a deployed commercial system reached a DME sensitivity of 26.5% (κ 0.38) on the same images on which its referable-DR κ was 0.81 (Duggal *et al.*, JMIR Med Inform 2025).

IDRiD defines its DME risk grade by the distance from the fovea to the nearest hard exudate: grade 0, no exudate; grade 1, exudates more than one disc diameter from the fovea; grade 2, within one disc diameter. Because the definition is geometric, a grade derived from the pixel-level exudate masks that IDRiD releases for 81 images, with its fovea and optic-disc annotations, should reproduce the expert grade if the grade is what its definition says; the test was committed before it was computed (Table S1). Of the 81 images, 54 could be linked to their grading-set copies (minimum pixel correlation 0.9999). The derived grade matched the expert grade in 96.3% of images, against

92.6% for a constant predictor, because almost every segmentation image is grade 2. On the 4 informative images (expert grade 0 or 1) the derivation failed twice, each time on a single small exudate lying just inside the threshold, so the pre-specified gate failed and no derived DME labels were used.

Supplementary Note 3. A per-grade verdict that depends on the cut-point procedure

A development source control (E12) compared runs identical except for whether Messidor-2 images came from the default or the native-resolution files, on the 953 development images both could use. Moderate recall, native minus default, under four ways of setting both arms' cut-points:

Cut-points for both arms	Moderate recall difference (points) [95% CI]
Shipped (0.5)	−6.9 [−12.1, −2.1]
Fitted on all rows (in-sample)	−10.8 [−16.2, −5.4]
Random two-fold cross-fit, grid 80	−6.6 [−12.0, −1.1]
Fold-wise cross-fit, grid 60 (development standard)	+0.4 [−5.7, +6.1]

The originally archived value, cited in the project as a difference that *survived* matched cut-points, is reproduced by the random-half procedure; under the fold-wise procedure used for every other development comparison it dissolves. A matched-decision-rule claim must therefore name its fitting procedure, and per-grade claims should be reported under at least two.

Supplementary Note 4. Correction of an archived ordinal-head comparison

Regenerating every development number from the archived per-image logits with independent code reproduced each archived matched QWK exactly except one. The archived comparison of the CORN head against the reference reported −0.0479 [−0.0656, −0.0325], with matched QWK 0.8168 for CORN. The archived script had decoded CORN's conditional logits as if they were cumulative; decoding them by cumulative product gives 0.8564 and a difference of −0.0082 [−0.0207, +0.0036], indistinguishable from the reference. The pre-specified verdict (no improvement) is unchanged; the claim that CORN is worse is withdrawn.

Supplementary Note 5. Seed variation (O1–O3)

The reference configuration was trained with 5 seeds on identical data, split and recipe (O1, the noise floor): matched development QWK 0.8646–0.8764 (s.d. 0.0043); APTOS QWK at the shipped rule 0.8900–0.8968 (s.d. 0.0026). Of 50 per-grade recall differences between pairs of seeds on APTOS, 15 had an interval excluding zero at the shipped rule (largest 8.2 points) and 29 with refitted cut-points (O2 held); on DDR 30 and 35 of 50. Mild recall on APTOS was 5.4–6.2% at the shipped rule and 63.8–72.7% under R4m in every seed (O3 held).

Table S13: Seed variation of the reference configuration (O1–O3). Five independent training seeds of E08 (identical data, split and recipe).

Seed	Dev QWK (matched)	APTOS QWK (R1)	APTOS mild R1 (%)	APTOS mild R4m (%)	DDR QWK (R1)
E08	0.8646	0.897	5.9	70.8	0.733
S1E08	0.8685	0.896	5.4	70.3	0.717
S2E08	0.8714	0.895	6.2	72.7	0.704
S3E08	0.8764	0.890	5.9	72.4	0.728
S4E08	0.8692	0.894	5.9	63.8	0.713

Supplementary Note 6. Reverse direction (O4–O6)

The reference recipe was trained on APTOS 2019 (own frozen 5-fold split, near-duplicates grouped) and evaluated on Messidor-2 and IDRiD. Matched out-of-fold QWK on APTOS was 0.9134. On Messidor-2 at the shipped rule mild recall was 34.4% (O4 was falsified); external cut-points raised it to 44.4% (O5 was falsified). The mean predicted-minus-true grade at the shipped rule was −0.26 (O6, under-grading, held).

Table S14: Reverse direction (O4–O6). The reference recipe trained on APTOS 2019 and evaluated on Messidor-2 and IDRiD.

Corpus	Rule	QWK	Mild recall (%)	Mild F1	Sens. (%)	Spec. (%)	Pred.−true grade
Messidor-2	R1	0.715	34.4	0.276	43.5	99.2	−0.26
	R2	0.705	16.3	0.168	47.3	98.9	−0.32
	R3	0.679	17.0	0.155	34.4	99.7	−0.36
	R4	0.779	25.9	0.273	77.9	93.8	−0.02
	R4m	0.755	44.4	0.325	82.7	91.7	+0.16
IDRiD	R1	0.807	64.0	0.239	81.7	96.9	−0.24
	R2	0.827	36.0	0.200	85.4	95.3	−0.23
	R3	0.810	44.0	0.173	75.5	97.9	−0.31
	R4	0.841	28.0	0.200	88.2	92.7	−0.04
	R4m	0.828	56.0	0.231	88.9	92.2	+0.08

Supplementary Note 7. Content-overlap audit

Every image of the development pool, the four external sets and EyePACS was reduced to a 256-bit difference hash after the training crop (APTOS and DDR in the first audit; DeepDRiD and SUSTech-SYSU, added in the second stage, in a second audit, post hoc). In this project's data, re-encodings of one photograph lie within 8 bits of each other and distinct photographs at least 28 bits apart; all pairs within 16 bits were counted and every pair within 8 bits was archived.

Development pool and external sets. No development image lay within 8 bits of any APTOS or DDR image (closest pairs 12 and 14 bits), and none lay within 16 bits of any DeepDRiD or SUSTech-SYSU image.

EyePACS and external sets. No external image was hash-identical to an EyePACS image. 18 APTOS–EyePACS pairs (11 APTOS images) and 2,304 DDR–EyePACS pairs (202 DDR images) and 55 DeepDRiD–EyePACS pairs (13 DeepDRiD images) lay within 8 bits, the closest at 6, 3 and 6 bits; no SUSTech-SYSU image came within 16 bits. The distances rise towards the threshold rather than peaking at zero, as expected of hash collisions between dark, low-contrast photographs. Removing every flagged image changed no development or EyePACS model's shipped-rule referable sensitivity, mild recall, QWK or referral AUC by more than 0.69 percentage points on DDR, 0.04 on APTOS and 0.15 on DeepDRiD.

Within corpora. The development pool contains 10 hash-identical pairs, 7 of which straddle two folds of the frozen split (14 images, missed when the split was built); removing them changes every model's matched development QWK by at most 0.0008. APTOS and DDR contain 148 and 120 hash-identical pairs.

Table S15: Content-overlap audit. Pairs of images across corpora whose 256-bit difference hashes lie within the stated Hamming distance.

Corpus pair	Images	Pairs ≤16 bits	≤8 bits	0 bits
within Development pool	2,260	29	10	10
Development pool vs APTOS	2,260 / 3,662	47	0	0
within APTOS	3,662	1,355	149	148
Development pool vs DDR	2,260 / 12,522	14	0	0
within DDR	12,522	23,021	265	120
APTOS vs DDR	3,662 / 12,522	1,361	1	0
EyePACS vs APTOS	35,126 / 3,662	10,086	18	0
EyePACS vs DDR	35,126 / 12,522	267,402	2,304	0
EyePACS vs Development pool	35,126 / 2,260	178	0	0
Threshold 16 bits; development pairs split across folds: 7				

Full decision-rule matrices

Table S16: Full decision-rule matrix, APTOS. QWK, accuracy, per-grade recall (%) and referable-DR sensitivity/specificity (%) of every model under every rule. R4/R4m are fitted on the external labels by two-fold cross-fitting and are diagnostic bounds, not performance.

Model	Rule	QWK	Acc.	No DR	Mild	Mod.	Sev.	PDR	Sens.	Spec.
E08	R1	0.897	73.5	97.0	5.9	64.2	64.2	52.2	99.5	84.3
	R2	0.883	68.2	98.3	8.9	39.0	56.0	65.4	99.1	85.1
	R3	0.882	67.9	98.3	4.6	39.2	56.0	65.4	99.3	84.2
	R4	0.904	78.2	98.8	18.1	77.9	49.7	47.8	96.6	87.0
	R4m	0.886	75.2	96.6	70.8	49.0	47.2	56.3	80.4	95.4
E09	R1	0.879	69.3	88.5	4.3	66.8	62.7	45.8	99.3	84.1
	R2	0.863	62.0	88.5	22.7	30.3	62.2	56.6	96.3	87.4
	R3	0.847	61.0	88.5	1.4	34.3	62.2	56.6	99.7	79.7
	R4	0.898	76.7	98.6	10.0	75.5	48.7	48.5	98.6	85.9
	R4m	0.886	75.6	97.8	61.6	53.6	50.3	48.5	81.1	94.7
E10	R1	0.887	70.8	94.4	4.3	59.9	72.5	45.8	99.0	84.2
	R2	0.885	70.1	98.1	7.0	50.3	42.0	63.1	98.8	84.7
	R3	0.884	69.8	98.1	3.2	50.8	42.0	63.1	99.2	84.0
	R4	0.898	76.8	98.0	13.5	75.5	50.3	49.2	97.2	85.9
	R4m	0.876	73.0	97.8	77.8	36.6	50.8	52.5	72.6	97.0
E11FULL	R1	0.903	74.9	98.1	4.9	65.6	63.2	60.3	99.3	84.3
	R2	0.900	72.9	98.9	7.6	54.5	63.2	64.1	99.0	84.9
	R3	0.900	72.8	98.9	7.0	54.6	63.2	64.1	99.1	84.8
	R4	0.907	79.1	98.6	16.8	81.8	39.4	54.6	96.5	86.6
	R4m	0.890	73.4	97.6	75.1	37.3	51.3	59.7	75.3	96.4
E14MAC	R1	0.880	68.7	95.5	5.7	53.9	72.5	31.9	98.7	84.3
	R2	0.879	67.2	98.6	5.7	39.6	72.0	42.7	98.3	84.7
	R3	0.877	67.1	98.6	1.4	40.9	72.0	42.7	99.2	83.6
	R4	0.889	74.6	98.9	10.3	69.0	52.8	40.3	97.6	85.7
	R4m	0.857	71.5	97.4	75.7	35.7	44.0	46.8	71.2	96.6
E15LPFT	R1	0.874	65.3	89.6	5.1	45.1	70.5	57.6	99.2	84.2
	R2	0.880	69.2	97.2	10.3	43.5	57.0	66.1	98.3	85.3
	R3	0.878	68.9	97.2	5.1	44.5	57.0	66.1	99.1	84.2
	R4	0.898	77.3	98.6	11.9	77.5	44.6	49.2	97.3	86.0
	R4m	0.878	75.5	97.7	70.3	49.3	34.2	61.4	78.5	96.0
E17NAT	R1	0.891	71.1	96.1	5.9	56.8	72.0	47.8	99.2	84.6
	R2	0.872	65.5	97.3	9.5	32.2	48.2	64.7	99.0	85.0
	R3	0.870	64.9	97.3	3.2	32.6	48.2	64.7	99.4	83.7
	R4	0.894	76.5	99.1	16.2	72.0	48.7	47.8	96.3	86.9
	R4m	0.874	73.6	97.7	72.7	41.7	50.3	51.2	74.0	96.3
E20CORAL	R1	0.879	66.3	96.1	14.3	39.1	64.8	42.4	98.2	85.3
	R2	0.869	65.7	98.1	13.8	33.6	47.7	53.2	98.2	85.4
	R3	0.867	65.2	98.1	4.1	35.2	47.7	53.2	99.4	83.3
	R4	0.887	75.2	97.8	15.7	73.8	35.8	42.4	97.1	85.7
	R4m	0.872	73.1	97.3	72.4	45.4	48.2	35.6	79.4	95.5
E22CORN	R1	0.892	72.7	97.6	1.6	62.3	71.0	45.8	99.3	83.9
	R2	0.889	69.3	97.0	10.8	47.3	76.2	43.1	98.4	85.3
	R3	0.887	69.1	97.0	5.4	48.5	76.2	43.1	99.3	84.0
	R4	0.896	77.2	98.2	14.6	76.3	54.4	44.7	96.9	86.4
	R4m	0.875	75.0	96.8	77.0	48.9	43.0	47.8	74.1	96.7
E21CNXA	R1	0.862	62.9	84.7	8.9	48.5	69.4	41.7	98.7	84.5
	R2	0.860	62.0	96.0	13.5	21.4	62.7	52.2	97.5	85.8
	R3	0.855	61.3	96.0	1.4	23.4	62.7	52.2	99.2	82.3
	R4	0.890	76.1	97.8	18.6	74.4	51.3	37.3	96.8	87.1
	R4m	0.873	74.3	98.0	67.0	48.3	51.8	41.0	79.6	95.3

Table S17: Full decision-rule matrix, DDR. QWK, accuracy, per-grade recall (%) and referable-DR sensitivity/specificity (%) of every model under every rule. R4/R4m are fitted on the external labels by two-fold cross-fitting and are diagnostic bounds, not performance.

Model	Rule	QWK	Acc.	No DR	Mild	Mod.	Sev.	PDR	Sens.	Spec.
E08	R1	0.733	73.7	97.7	33.3	51.2	58.5	50.3	65.5	98.7
	R2	0.734	68.9	98.8	22.1	38.2	53.4	50.3	62.5	99.0
	R3	0.742	70.5	98.8	17.5	43.4	53.4	50.3	67.1	98.3
	R4	0.805	69.8	86.2	25.2	57.8	57.2	49.2	80.8	93.2
	R4m	0.787	67.0	83.6	59.8	49.8	55.9	45.3	66.8	98.4
E09	R1	0.697	68.8	93.5	20.5	46.4	54.7	46.3	63.9	97.3
	R2	0.708	58.6	92.0	31.7	18.1	56.8	47.0	50.4	99.2
	R3	0.733	66.5	92.0	11.7	43.0	56.8	47.0	72.1	93.4
	R4	0.753	60.7	77.1	38.4	43.6	48.3	50.4	70.9	94.1
	R4m	0.744	57.3	71.6	51.6	41.2	61.0	41.4	64.9	96.9
E10	R1	0.737	70.5	96.1	43.7	44.8	59.3	42.7	62.8	98.2
	R2	0.740	69.3	97.4	27.1	41.9	41.9	47.8	63.4	98.2
	R3	0.748	71.1	97.4	21.6	47.6	41.9	47.8	68.3	97.4
	R4	0.793	67.2	85.1	23.5	53.1	67.4	43.3	80.4	92.6
	R4m	0.775	63.4	83.0	57.9	40.9	58.5	43.4	66.7	97.6
E11FULL	R1	0.755	76.1	98.2	29.8	56.7	55.5	56.1	70.3	98.7
	R2	0.756	72.5	98.9	22.1	46.9	56.4	55.6	65.2	99.0
	R3	0.757	72.6	98.9	21.4	47.4	56.4	55.6	65.6	98.9
	R4	0.813	72.8	89.2	24.6	60.2	60.6	58.4	81.2	94.2
	R4m	0.799	66.7	84.2	58.7	46.3	64.8	53.1	69.4	98.4
E14MAC	R1	0.736	70.2	96.3	42.1	43.7	71.2	39.6	61.7	98.9
	R2	0.729	68.1	98.6	19.5	37.1	62.7	45.2	62.5	98.8
	R3	0.740	70.4	98.6	10.2	44.9	62.7	45.2	69.4	97.4
	R4	0.793	67.8	84.7	24.0	55.2	61.9	45.5	80.1	93.1
	R4m	0.776	64.8	84.4	57.1	43.8	68.2	38.4	65.6	98.3
E15LPFT	R1	0.764	69.9	92.5	45.2	43.7	63.1	62.3	66.1	98.3
	R2	0.743	68.9	97.5	21.4	38.2	51.7	60.0	62.2	98.8
	R3	0.753	71.2	97.5	16.5	45.4	51.7	60.0	68.4	97.9
	R4	0.794	67.5	84.6	35.2	52.0	66.5	48.6	76.8	94.5
	R4m	0.786	64.4	81.0	57.6	44.5	61.0	54.0	67.6	97.9
E17NAT	R1	0.743	71.5	96.9	38.9	46.4	58.9	45.5	64.6	98.2
	R2	0.765	66.9	96.1	33.8	35.0	54.7	48.5	65.2	98.3
	R3	0.772	68.9	96.1	26.5	41.6	54.7	48.5	71.0	97.0
	R4	0.801	67.5	85.7	31.4	51.8	55.5	48.2	79.4	93.4
	R4m	0.789	63.4	83.3	62.1	40.4	63.6	41.1	67.5	97.8
E20CORAL	R1	0.740	65.2	93.8	37.8	36.0	59.3	33.1	58.9	98.5
	R2	0.728	65.6	96.2	24.9	33.8	50.8	43.8	58.9	98.5
	R3	0.743	69.6	96.2	14.4	46.5	50.8	43.8	69.7	96.6
	R4	0.778	64.2	82.9	37.1	46.4	57.6	43.5	75.1	94.1
	R4m	0.768	60.6	81.0	58.7	37.1	64.4	35.5	65.6	97.7
E22CORN	R1	0.699	72.4	98.7	25.9	49.1	56.4	42.5	63.8	98.6
	R2	0.707	66.5	98.0	32.4	34.7	64.0	30.0	54.7	99.3
	R3	0.730	70.2	98.0	24.0	46.3	64.0	30.0	65.0	98.3
	R4	0.785	67.4	85.2	22.7	54.5	56.4	41.9	79.4	92.8
	R4m	0.766	63.6	85.9	56.5	39.1	63.1	36.3	63.2	98.6
E21CNXA	R1	0.710	65.4	88.0	34.9	45.0	51.7	34.7	62.6	98.4
	R2	0.701	62.1	94.9	23.8	26.3	53.8	40.4	55.1	99.2
	R3	0.726	67.8	94.9	8.7	44.5	53.8	40.4	71.3	95.8
	R4	0.747	58.7	78.5	41.4	37.3	56.4	40.3	70.0	96.3
	R4m	0.738	56.9	70.5	53.5	43.0	59.3	32.5	67.4	97.3

Table S18: Full decision-rule matrix, DeepDRiD. QWK, accuracy, per-grade recall (%) and referable-DR sensitivity/specificity (%) of every model under every rule. R4/R4m are fitted on the external labels by two-fold cross-fitting and are diagnostic bounds, not performance.

Model	Rule	QWK	Acc.	No DR	Mild	Mod.	Sev.	PDR	Sens.	Spec.
E08	R1	0.792	64.3	90.8	19.8	75.4	19.8	37.5	86.6	93.9
	R2	0.812	64.0	93.5	10.8	58.5	35.9	37.5	83.7	94.7
	R3	0.815	64.5	93.5	6.3	63.3	35.9	37.5	86.6	93.4
	R4	0.833	60.1	80.6	33.8	49.5	44.1	33.0	86.9	93.0
	R4m	0.818	61.5	73.0	40.5	64.1	58.2	10.7	87.0	92.4
E09	R1	0.747	60.8	92.3	10.8	57.3	22.9	34.8	77.8	95.1
	R2	0.757	57.2	89.6	22.5	23.9	39.5	35.7	63.1	97.9
	R3	0.777	61.6	89.6	7.2	54.3	39.5	35.7	83.1	90.4
	R4	0.793	57.9	81.2	31.5	28.1	55.9	32.1	76.5	95.2
	R4m	0.781	57.0	75.3	32.0	46.7	45.5	29.5	77.1	93.1
E10	R1	0.797	63.4	87.9	32.0	66.8	24.6	37.5	83.8	93.9
	R2	0.808	63.4	90.6	18.5	62.8	29.9	39.3	83.9	93.9
	R3	0.813	63.7	90.6	12.6	67.6	29.9	39.3	87.7	92.0
	R4	0.820	59.7	78.7	35.6	54.3	38.1	40.2	86.7	92.3
	R4m	0.806	56.5	63.2	55.4	54.8	51.1	27.7	83.8	93.9
E11FULL	R1	0.819	66.5	90.7	23.0	81.7	19.8	48.2	90.3	92.3
	R2	0.828	65.5	91.6	18.9	71.9	26.3	45.5	87.6	93.1
	R3	0.830	65.5	91.6	18.9	72.4	26.3	45.5	88.1	93.1
	R4	0.844	66.3	83.2	31.5	65.6	57.9	26.8	91.4	91.1
	R4m	0.839	61.8	79.2	41.0	65.3	30.8	45.5	87.8	93.0
E14MAC	R1	0.777	64.3	89.6	28.8	67.1	31.1	24.1	81.1	95.6
	R2	0.788	64.8	94.3	11.3	58.5	41.8	24.1	81.7	95.9
	R3	0.791	65.3	94.3	5.4	64.6	41.8	24.1	86.6	92.7
	R4	0.805	58.6	80.7	34.2	39.7	48.0	26.8	83.0	94.2
	R4m	0.797	59.9	75.9	40.1	59.5	42.9	23.2	84.3	94.0
E15LPFT	R1	0.794	63.9	88.8	24.8	62.8	30.5	48.2	85.1	92.9
	R2	0.799	63.0	92.3	11.7	61.3	27.7	43.8	83.2	93.8
	R3	0.801	63.6	92.3	8.6	65.8	27.7	43.8	86.1	92.3
	R4	0.822	62.8	84.0	26.1	54.0	48.9	37.5	87.5	90.7
	R4m	0.809	57.8	73.0	43.7	57.5	31.9	44.6	84.6	92.8
E17NAT	R1	0.797	63.2	86.0	34.2	67.6	26.3	36.6	84.7	93.5
	R2	0.826	63.1	85.6	29.3	55.0	43.2	39.3	86.1	93.0
	R3	0.823	63.0	85.6	18.9	60.1	43.2	39.3	89.1	89.6
	R4	0.822	59.5	79.5	36.9	41.5	48.3	39.3	83.1	94.0
	R4m	0.816	57.4	71.8	42.8	53.0	39.0	42.0	85.4	93.3
E20CORAL	R1	0.798	63.2	89.6	25.2	52.8	44.1	20.5	78.7	95.6
	R2	0.797	63.5	92.6	18.9	49.2	43.2	29.5	78.7	95.6
	R3	0.803	65.4	92.6	9.9	63.8	43.2	29.5	86.9	92.3
	R4	0.823	60.4	82.8	28.8	35.2	59.3	32.1	83.7	92.9
	R4m	0.795	53.7	55.8	60.4	50.5	55.9	27.7	84.6	93.5
E22CORN	R1	0.770	64.8	91.9	14.9	73.1	26.0	35.7	85.2	92.6
	R2	0.794	65.0	90.2	23.9	58.3	45.5	27.7	78.9	95.7
	R3	0.798	65.8	90.2	14.9	67.1	45.5	27.7	85.1	91.9
	R4	0.818	62.4	83.0	37.4	46.5	51.1	34.8	82.3	94.2
	R4m	0.813	62.2	75.8	42.3	59.5	52.8	29.5	83.0	93.8
E21CNXA	R1	0.778	61.9	89.3	24.8	60.1	27.1	27.7	80.8	94.6
	R2	0.790	61.6	92.6	18.5	38.7	43.2	33.9	75.8	96.4
	R3	0.798	63.6	92.6	7.7	55.0	43.2	33.9	86.5	91.5
	R4	0.811	58.5	80.7	27.5	39.2	51.7	27.7	86.6	91.0
	R4m	0.796	60.0	72.2	44.1	56.0	58.5	10.7	82.9	93.6

Table S19: Full decision-rule matrix, SUSTech. QWK, accuracy, per-grade recall (%) and referable-DR sensitivity/specificity (%) of every model under every rule. R4/R4m are fitted on the external labels by two-fold cross-fitting and are diagnostic bounds, not performance.

Model	Rule	QWK	Acc.	No DR	Mild	Mod.	Sev.	PDR	Sens.	Spec.
E08	R1	0.923	82.0	96.4	33.3	64.7	94.5	39.7	99.0	97.9
	R2	0.904	74.2	97.5	25.0	43.0	53.4	63.8	98.2	98.3
	R3	0.902	74.4	97.5	12.5	44.4	53.4	63.8	99.2	97.3
	R4	0.932	84.4	98.1	25.0	73.4	90.4	22.4	97.4	98.6
	R4m	0.900	76.7	82.9	79.2	70.4	82.2	41.4	92.5	99.5
E09	R1	0.892	78.6	90.6	29.2	64.9	87.7	43.1	99.2	93.9
	R2	0.890	69.5	88.7	58.3	37.8	74.0	58.6	97.4	98.6
	R3	0.857	69.5	88.7	12.5	40.8	74.0	58.6	99.6	89.3
	R4	0.925	85.2	96.2	20.8	80.5	87.7	19.0	97.4	98.5
	R4m	0.912	81.2	89.5	50.0	75.3	71.2	53.4	97.4	98.3
E10	R1	0.929	80.5	97.8	50.0	55.3	97.3	41.4	98.6	98.8
	R2	0.901	76.2	99.5	33.3	50.1	26.0	67.2	98.2	98.8
	R3	0.903	76.5	99.5	33.3	51.0	26.0	67.2	99.0	98.8
	R4	0.937	87.5	99.4	25.0	84.9	71.2	20.7	99.2	98.5
	R4m	0.918	81.5	95.7	75.0	62.2	84.9	46.6	90.1	99.7
E11FULL	R1	0.929	83.2	98.1	33.3	64.9	94.5	43.1	98.8	98.5
	R2	0.924	79.7	99.2	20.8	54.2	79.5	51.7	98.4	98.9
	R3	0.924	79.7	99.2	16.7	54.5	79.5	51.7	98.6	98.8
	R4	0.934	87.3	98.4	20.8	83.3	80.8	27.6	98.4	98.6
	R4m	0.919	81.4	92.9	62.5	69.9	78.1	41.4	95.8	99.8
E14MAC	R1	0.919	77.3	95.7	58.3	51.5	95.9	24.1	98.2	99.2
	R2	0.916	75.6	99.7	20.8	40.5	86.3	43.1	98.2	99.1
	R3	0.917	75.5	99.7	4.2	41.4	86.3	43.1	99.0	98.3
	R4	0.928	84.3	99.2	20.8	74.0	78.1	20.7	97.6	99.1
	R4m	0.916	80.0	89.9	75.0	71.5	74.0	36.2	96.6	99.7
E15LPFT	R1	0.910	75.2	93.2	41.7	47.9	82.2	56.9	98.4	98.2
	R2	0.906	75.8	98.3	20.8	44.7	60.3	70.7	97.4	98.9
	R3	0.902	76.1	98.3	8.3	46.3	60.3	70.7	98.6	97.4
	R4	0.936	85.2	98.1	20.8	77.0	74.0	37.9	98.0	98.2
	R4m	0.920	79.6	90.6	75.0	65.2	75.3	56.9	95.8	99.7
E17NAT	R1	0.929	80.1	97.9	37.5	54.2	98.6	43.1	99.0	98.8
	R2	0.895	70.2	98.7	29.2	32.9	26.0	67.2	98.0	98.8
	R3	0.897	70.7	98.7	20.8	35.1	26.0	67.2	99.8	98.3
	R4	0.937	87.1	99.4	12.5	81.1	83.6	27.6	99.8	98.3
	R4m	0.917	80.4	91.9	58.3	66.6	76.7	55.2	95.4	99.2
E20CORAL	R1	0.909	71.9	93.3	50.0	41.1	71.2	43.1	97.6	98.9
	R2	0.902	72.1	97.5	41.7	38.4	37.0	65.5	97.6	98.9
	R3	0.899	72.4	97.5	16.7	40.8	37.0	65.5	99.4	97.1
	R4	0.934	85.3	98.3	8.3	81.6	72.6	15.5	98.8	98.3
	R4m	0.887	70.9	73.9	70.8	74.0	58.9	34.5	95.8	99.5
E22CORN	R1	0.920	82.9	97.6	16.7	66.3	91.8	43.1	99.2	97.1
	R2	0.919	77.4	96.2	29.2	50.7	91.8	43.1	97.6	98.5
	R3	0.919	77.7	96.2	12.5	52.6	91.8	43.1	99.2	97.3
	R4	0.929	85.6	97.5	8.3	81.4	86.3	13.8	98.0	98.3
	R4m	0.906	78.0	84.8	70.8	72.6	71.2	50.0	95.2	99.5
E21CNXA	R1	0.883	65.3	77.7	41.7	46.0	80.8	43.1	98.8	98.0
	R2	0.889	65.6	91.9	37.5	26.0	49.3	60.3	98.0	99.1
	R3	0.876	65.9	91.9	12.5	28.8	49.3	60.3	100.0	94.7
	R4	0.926	83.3	98.4	4.2	76.2	76.7	5.2	98.4	98.8
	R4m	0.900	74.2	88.1	62.5	58.1	60.3	46.6	94.4	99.7